

Nutrient Deficiency Detection in Mobile Captured Guava Plants using Light Weight Deep Convolutional Neural Networks

Sona Haris¹, Keshav Shesha Sai¹, N. Shobha Rani^{1*}, Pushpa B. R¹

Department of Computer Science, Amrita School of Computing, Mysuru Campus

Amrita Vishwa Vidyapeetham, India

Email: n_shobharani@my.amrita.edu

Abstract—Nutrition deficiency in plants is a major problem that affects their growth, yield, and nutritional value. Over the past few years, there has been a significant growth in the application of machine learning and computer vision techniques, in early detection and classification of plant disorders. This study, proposes a deep learning-based approach for detecting nutritional deficiencies in guava leaf images. A dataset of guava leaf images captured using mobile devices, containing various nutritional deficiencies including magnesium and phosphorous, was acquired for training the model. A pre-trained deep CNN model is employed to extract convolved features and detect the affected regions, categorizing them as nutritional deficient or non-nutritional deficient. Experimental results show that the proposed method achieved an accuracy of 87% in detecting nutritional deficiencies in guava leaf images. These outcomes demonstrate that the proposed approach provides a reliable and accurate method for early detection of nutritional deficiencies in guava leaves. This approach has the potential to be deployed in the agricultural domain for the effective diagnosis of plant nutrient deficiencies, ultimately increasing crop productivity and nutritional quality.

Keywords— *Convolutional Neural Network, Nutritional Deficiencies Detection, Classification*

I. INTRODUCTION

Nutritional deficiency in guava plants often has significant consequences in their development, productivity, and fruit quality. These deficiencies can result stunted growth, leaf yellowing or purpling, reduced fruit yield, and increased vulnerability to pests and diseases. Guava is one of the widely cultivated and grown tropical and subtropical fruit. *Psidium guajava* (L.) belongs to the Myrtaceae family and it is an important fruit in tropical areas like India, Indonesia, Pakistan, Bangladesh, and South America. The changes occurring in plant leaf reflect the plant health. The leaves of guava are usually dull green in color and leathery texture. The most obvious visual symptom of a plant nutrient deficiency is leaf discoloration or distortion which is also seen in guava leaf where leaf color changes due to nutrient deficiency present in them.

Manual nutrient deficiency detection in guava plants is a time-consuming and labor-intensive procedure that requires expert knowledge and expertise. As a result, there's a demand for fully automated system that can detect nutritional deficiencies in guava plants accurately and efficiently. There are a number of current models for creating a system to identify nutritional deficiencies in images of plants, with deep learning-based approaches

being the most popular and having the highest accuracy rates. CNN has shown a great performance in image classification tasks as it is designed to handle complex variability of images. However, further study is required to increase the robustness and scalability of these models for actual crop monitoring.

The current approaches still have certain limitations. The accuracy of the models is mainly dependent on the quality and number of training samples. In some circumstances, obtaining a diverse and relevant dataset may be challenging, resulting in overfitting or incorrect predictions. Several plant diseases may have visually similar symptoms, making accurate identification difficult. The models may struggle to detect disorders that are not noticeable to the human eye or that are still in their early stages. Other factors such as lighting conditions, camera angle, and image resolution may have an impact on model performance.

The proposed system is based on Convolutional Neural Network. Designing a CNN-based system for detecting nutritional deficiencies in guava plants can provide a cost-effective and scalable solution to this challenge. The system uses guava leaf images captured using mobile with various nutrient deficiency. The main objectives of the project are:

1. To collect and preprocess the images of guava leaf.
2. To devise and train the CNN model for classification of guava leaves into different categories based on the type of deficiency present in them.
3. Evaluation of the performance of the CNN model using various performance metrics.

II. LITERATURE SURVEY

The use of CNNs has quickly emerged as a favored approach for identifying diseases in plants [1]. Savita N et al [2] studied different methods for identifying plant leaf diseases based on their physical characteristics. The evaluated methods like K-Nearest Neighbor, neural networks, fuzzy logic, and evolutionary algorithms, and concluded that support vector machine (SVM) was the best for handling large, high-dimensional datasets. However, SVM implementation is challenging due to selecting the correct parameter for training nonlinear data, which is solved through quadratic optimization.

In their study of tomato leaf disorders using the plant village dataset, Vijay et al [3] employed both CNN and K-

nearest neighbor (KNN) models for classification. LIME is utilized to provide interpretability for the models' predictions. The results showed that the CNN model outperformed the KNN model in detecting leaf disease, achieving an accuracy of 98.5% compared to the KNN model's accuracy of only 83.6%.

According to Prajwala et al [4], a significant limitation of traditional techniques for disease detection is the requirement to manually extract features from data. In contrast, CNNs leverage convolutional operations to automatically learn features from input data, making them particularly adept at identifying visual patterns. Consequently, CNNs have been employed for diagnosing and treating plant diseases.

Using a Support Vector Machine (SVM) classifier, Arivazhagan and team [5] suggested the utilization of textural features to classify hazardous regions across ten plant species, resulting in a 94.74% accuracy rate. Agarwal et al. [6] utilized the Plant Village dataset to develop a CNN model for tomato crops with 91% accuracy. The model underwent data augmentation and was compared with a pre-trained model. Experimental results showed an average precision of 91.2%, and a testing accuracy range of 76% to 100% for different classes. The proposed model required significantly less storage compared to pre-trained models at about 1.5 MB, highlighting its advantages.

In order to identify the species of plant, Han et al. [7] suggested a deep learning method that extracted distinctive features from images of plant leaves. Their method used CNN feature learning, which outperformed manually generated features with a 99.6% accuracy rate. The study also found that the venation structure is crucial in distinguishing between different plant species. Furthermore, a DN visualization tool was utilized to validate the network's internal operations and behaviour.

Kadir et al. [8] introduced an algorithm for classifying leaf images that consists of extracting leaf features using a feature extractor, processing the features using a PNN, and segmenting the leaf from the background. Segmentation is carried out by creating a 20-bin intensity histogram of the image, identifying the bin with the lowest value between the two main peaks for the leaf and background, and using its median as a threshold to separate the leaf from its surroundings. The algorithm is applied for categorizing leaf images.

Using hue, Patil et al. [9] created a method to recognise sick tomato leaves. Low order moments like Mean, Standard Deviation, and Skewness were used to measure the distribution of color in the research, which concentrated on identifying three prevalent diseases affecting tomato plants. The research on ten different tomato leaves infected with various diseases was implemented using MATLAB image processing and statistical tools, yielding nine color moment characteristics for each image.

Lewis et al. [10] conducted a study to classify and detect nutritional deficiencies in coffee plants using image processing and Convolutional Neural Network (CNN). The results of the study showed that the proposed CNN model

had a high accuracy in detecting and classifying the nutritional deficiencies in coffee plants.

Ali et al. [11] introduced a customized CNN (Convolutional Neural Network) model for identifying and categorizing nutritional deficiency grape leaves. Sathyavani [12] proposed an approach that involves recognizing patterns from leaf images captured by IoT devices and a nutrition analyser device using a Convolutional Neural Network (CNN). To address issues such as pixel loss, mislabelling, noise, imbalance, and large size in the data, the SMOTE approach was utilized, which significantly improved the accuracy of the CNN classifier.

The majority of these studies have used machine learning approaches and computer vision techniques, to develop automated system agriculture and plant health monitoring. These systems have a significant impact on improving the productivity of agriculture.

III. DATASET

The dataset used in this study consists of guava leaf images with various nutrition deficiencies, including nitrogen, phosphorus, magnesium, and manganese deficiency. The dataset was collected from a guava farm located in the Indian state of Karnataka. The images are captured using smartphones. A total of 664 guava leaf images were collected from different guava trees and they are assigned to either the deficient or non-deficient category. The photos are labelled with information regarding the specific nutritional deficiency that is exhibited in them.

Table I
NUMBER OF SAMPLES COLLECTED FOR EACH CLASS.

class	symptoms	Number of images
Phosphorous	slow, weak, and stunted growth and may result in dark to blue-green discoloration on older leaves. In severe cases, affected leaves and stems may turn purple."	424
Magnesium	The leaf tissue in between the veins might be yellow, copper, or reddish, but the veins on the leaf are always green.	108
Manganese	Symptoms first appear as chlorosis in young tissues. Mn chlorosis shows up as tiny yellow spots.	74
Healthy	Healthy guava leaves are typically dark green in color.	58
Total number of samples		664

IV. METHODOLOGY

Image Acquisition: The dataset consists of images of guava leaves that have been captured using a mobile device like Redmi and One plus. The leaves are affected by various deficiencies such as phosphorus, magnesium, and manganese. The images have been captured under different lighting conditions, and at different stages of the deficiency. The leaves were randomly selected from different trees and angles to ensure the diversity of the dataset. Each image is in JPEG format and has a resolution of 1280x720 pixels.

Image Transformation: A transformation approach is used to pre-process image data for training deep learning models. This pipeline consists of scaling input images to a fixed size of 150x150 pixels, randomly flipping them horizontally, turning them to PyTorch tensors, and normalizing the tensor values that range between 0 and 1. Subtraction of the mean and division by the standard deviation yields the standardized tensor values. These pre-processing methods are necessary as several deep learning models require fixed-sized input data, and standardization and normalization are often found to enhance model performance by enhancing the variance of the input image.

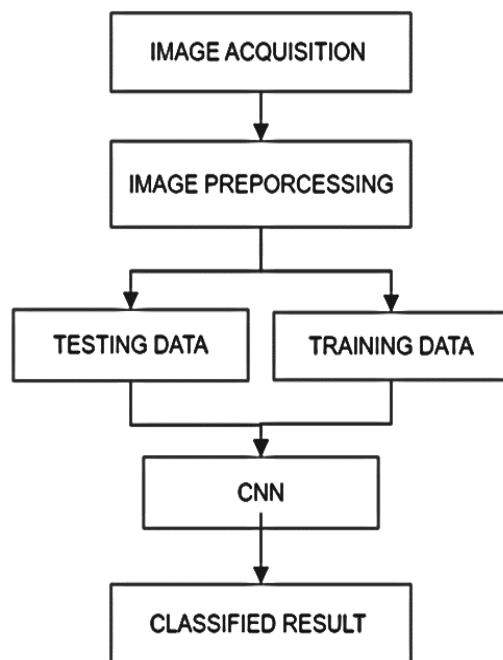


Fig. 2. General flow of the system

Convolutional neural network (CNN): A CNN is a deep learning model that is specifically designed to work with image data. The model consists of multiple layers, such as convolutional layers, pooling layers, and fully connected layers, that help to extract features from the input image and classify it into one or more categories. The fundamental components of a CNN are convolutional layers since they are responsible for understanding and extracting information from the input images. Convolutional layers in a CNN use a set of filters to apply to the input images and generate a set of output feature maps. Convolutional layers may learn to understand a variety of features and characteristics in the input image by

applying multiple filters parallelly. This is crucial for image classification and other computer vision tasks since various objects and situations may be identified by a combination of several visual characteristics.

Model evaluation: The performance of the model is measured as accuracy, precision, recall and fl_score on test dataset explained in equation (1)-(4).

$$\text{accuracy} = (T^+ + T^-) / (T^+ + T^- + F^+ + F^-) \quad (1)$$

$$\text{precision} = T^+ / (T^+ + F^+) \quad (2)$$

$$\text{recall} = T^+ / (T^+ + F^-) \quad (3)$$

$$\text{fl_score} = 2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall}) \quad (4)$$

where, T^+ is true positive values, T^- is true negative values, F^+ is false positive values and F^- is false negative values.

V. RESULT AND DISCUSSION

The CNN model was used to detect the nutrient deficiency in guava leaf which helps in accurate and efficient classification of images of guava leaves based on their deficiency. The CNN model is consist of three convolutional layers followed by fully connected layer using ReLU activation function.

The CNN model was trained on a dataset of images of guava leaves belonging to four different classes - magnesium deficient, manganese deficient, phosphorus deficient, and healthy leaves. The model was trained using the PyTorch framework.

Table II
TRAINING AND TESTING DATASET DISTRIBUTION

	Classes	Number of samples	Total
Training	Healthy	43	534
	Magnesium	78	
	Manganese	43	
	Phosphorous	370	
Testing	Healthy	15	130
	Magnesium	30	
	Manganese	31	
	Phosphorous	54	

The network consists of several layers including convolutional layers, batch normalization, max pooling, and a fully connected layer. The network takes 3-channel input images and produces a vector of length num_classes as output. The image transformation helps in increasing the amount of training dataset, standardize and pre-process the input image which helps in improving the accuracy and efficiency of the model as well as reduce overfitting issues. The convolutional layers detect local features in image and combine them to identify complex patterns in the image which help in finding highly discriminative features that are helpful in identifying various symptoms shown in leaves due to nutrient deficiency. The features extracted are various image representations such as edges, textures and shapes. Later these features are used for nutrient deficiency detection. During training, the model learns the appropriate

weights for each layer using backpropagation and gradient descent.

The model performs well overall, with an average accuracy of about 87% with individual class accuracies of 91.5% for healthy, 96.2% for magnesium, 86.9% for manganese, and 79.2% for phosphorus. This shows that the model can accurately classify the majority of the data across all classes.

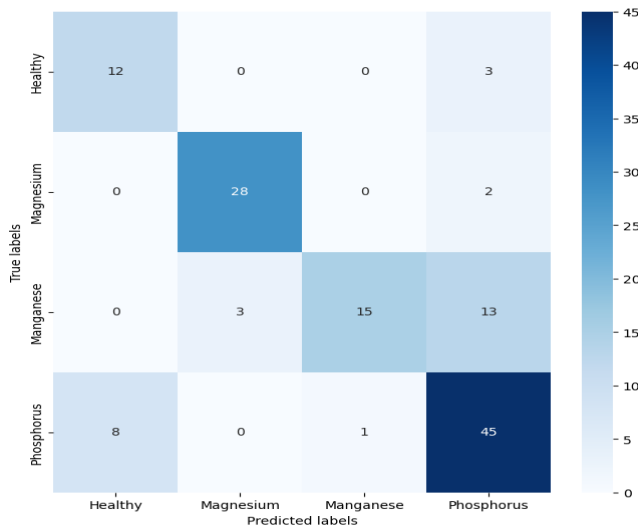


Fig. 5. Confusion matrix for the model

The provided values from confusion matrix were used to calculate various performance metrics such as accuracy, precision, recall and F1 score which helps in analyzing the performance of the model than an accuracy score in identifying healthy and leaves with nutrient deficiencies.

Table III
PERFORMANCE PARAMETERS

class label	accuracy	precision	recall	f1_score
Healthy	0.91538462	0.6	0.8	0.68571429
Magnesium	0.96153846	0.90322581	0.933333	0.91803279
Manganese	0.86923077	0.9375	0.483870	0.63829787
Phosphorus	0.79230769	0.71428571	0.833333	0.76923077

VI. CONCLUSION

The CNN model was trained to classify the guava leaf images based on their depicted deficiency symptoms, the

proposed system achieves high accuracy and performance. Furthermore, the model achieves an accuracy of 91.5%, 96.2% 86.9% and 79.2% for healthy, magnesium, manganese and phosphorus respectively. The system can also potentially be used in other crop species and can be expanded to address other plant health issues.

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